

Last login: Thu Jun 18 21:11:31 on ttys000
^[[A
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# SET ROLE alice;  
SET  
tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)  
SELECT COUNT(*)  
FROM l
```

```
SELECT l_orderkey, l_linenum  
FROM lineitem  
WHERE l_orderkey > 2000000  
) l_alias;
```

WARNING: Log 1: Planner Input:

```
DETAIL: (QUERY  
:commandType 1  
:querySource 0  
:canSetTag true  
:utilityStmt <>  
:resultRelation 0  
:hasAggs true  
:hasWindowFuncs false  
:hasTargetSRFs false  
:hasSubLinks false  
:hasDistinctOn false  
:hasRecursive false  
:hasModifyingCTE false  
:hasForUpdate false  
:hasRowSecurity true  
:hasGroupRTE false  
:isReturn false  
:cteList <>  
:rtable (  
  {RANGETBLENTY  
    :alias  
      {ALIAS  
        :aliasname l_alias  
        :colnames <>  
      }  
    :eref  
      {ALIAS  
        :aliasname l_alias  
        :colnames ("l_orderkey" "l_linenum")  
      }  
    :rtekind 1  
    :subquery  
      {QUERY  
        :commandType 1  
        :querySource 0  
        :canSetTag true  
        :utilityStmt <>  
        :resultRelation 0  
        :hasAggs false  
        :hasWindowFuncs false  
        :hasTargetSRFs false  
        :hasSubLinks true  
        :hasDistinctOn false  
        :hasRecursive false  
        :hasModifyingCTE false  
        :hasForUpdate false  
        :hasRowSecurity true  
        :hasGroupRTE false  
        :isReturn false  
        :cteList <>  
        :rtable (  
          {RANGETBLENTY  
            :alias <>  
            :eref  
              {ALIAS  
                :aliasname lineitem  
                :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum"  
                  "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl  
                    ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"  
                      "l_shipinstruct" "l_shipmode" "l_comment")  
              }  
            :rtekind 0  
            :relid 16566  
            :inh true  
            :relkind r  
            :rellockmode 1  
            :perminfoindex 1  
            :tablesample <>  
            :lateral false  
            :inFromCl true  
            :securityQuals (  
              {SUBLINK  
                :subLinkType 0  
                :subLinkId 0  
                :testexpr <>  
                :opernam <>  
                :subselect  
                  {QUERY  
                    :commandType 1  
                    :querySource 0  
                    :canSetTag true  
                    :utilityStmt <>  
                    :resultRelation 0  
                    :hasAggs false  
                    :hasWindowFuncs false  
                    :hasTargetSRFs false  
                    :hasSubLinks false  
                    :hasDistinctOn false  
                    :hasRecursive false  
                    :hasModifyingCTE false  
                    :hasForUpdate false  
                    :hasRowSecurity false  
                    :hasGroupRTE false  
                    :isReturn false  
                    :cteList <>  
                    :rtable (  
                      {RANGETBLENTY  
                        :alias  
                          {ALIAS  
                            :aliasname s  
                            :colnames <>  
                          }  
                        :eref  
                          {ALIAS  
                            :aliasname s  
                            :colnames ("s_suppkey" "s_name" "s_address" "s_nationk  
                              ey" "s_phone" "s_acctbal" "s_comment")  
                          }  
                        :rtekind 0  
                        :relid 16541  
                        :inh true  
                        :relkind r  
                        :rellockmode 1  
                        :perminfoindex 1  
                        :tablesample <>  
                        :lateral false  
                        :inFromCl true  
                        :securityQuals <>  
                      }  
                    }  
                  }  
                :rteperminfos (  
                  {RTEPERMISSIONINFO  
                    :relid 16541  
                    :inh true  
                    :requiredPerms 2  
                    :checkAsUser 0  
                    :selectedCols (b 8 13)  
                    :insertedCols (b)  
                    :updatedCols (b)  
                  }  
                )  
              }  
            :joinintree
```



```

:rtperminfos (
  {RTEPERMISSIONINFO
   :relid 16566
   :inh true
   :requiredPerms 2
   :checkAsUser 0
   :selectedCols (b 8 11)
   :insertedCols (b)
   :updatedCols (b)
  }
)
:jointree
{FROMEXPR
 :fromlist (
  {RANGETBLREF
   :rtindex 1
  }
 )
 :quals
 {OEXPR
  :opno 521
  :opfuncid 147
  :opresulttype 16
  :opreset false
  :opcollid 0
  :inputcollid 0
  :args (
   {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 123
   }
   {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 136
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
   }
  )
  :location 134
 }
 }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
 {TARGETENTRY
  :expr
  {VAR
   :varno 1
   :varattno 1
   :vartype 23
   :vartypmod -1
   :varcollid 0
   :varnullingrels (b)
   :varlevelsup 0
   :varreturningtype 0
   :varnosyn 1
   :varattnosyn 1
   :location 70
  }
  :resno 1
  :resname 1_orderkey
  :ressortgroupref 0
  :resorigtbl 16566
  :resorigcol 1
  :resjunk false
 }
 {TARGETENTRY
  :expr
  {VAR
   :varno 1
   :varattno 4
   :vartype 23
   :vartypmod -1
   :varcollid 0
   :varnullingrels (b)
   :varlevelsup 0
   :varreturningtype 0
   :varnosyn 1
   :varattnosyn 4
   :location 82
  }
  :resno 2
  :resname 1_linenummer
  :ressortgroupref 0
  :resorigtbl 16566
  :resorigcol 4
  :resjunk false
 }
 )
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:permInfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
)
:rtperminfos <>
:jointree
{FROMEXPR
 :fromlist (
  {RANGETBLREF
   :rtindex 1
  }
 )
 :quals <>
}
:mergeActionList <>
:mergeTargetRelation 0

```

```

:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 0
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggssplit 0
        :aggno -1
        :aggtransno -1
        :location 43
      }
    :resno 1
    :resname count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 propparallel=s (s=safe r=restricted u=unsafe)

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname s
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname s
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals (
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
          }
          {PARAM
            :paramkind 1
            :paramid 0
            :paramtype 23
            :paramtypmod -1
            :paramcollid 0
          }
        )
      }
    )
  }
}

```

```

        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
    }
    {CONST
        :consttype 701
        :consttypmod -1
        :constcollid 0
        :constlen 8
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
    }
)
}
:location -1
}
}
:mergeActionList <>
:mergeFargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :right_hasheqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0} :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in safe_list=0 + reason=outer reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :right_hasheqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0} :right_hasheqoperator 0})
= false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

```

```

WARNING: Log 3: At the starting of Plan creation:
DETAIL:
{PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe false
:parallel_workers 0
:rows 1
:disabled_nodes 0
:startup_cost 0
:total_cost 368
:pathkeys <>
}

```

```

WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGELEFTENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
)
}

```

```

:eref
{
  ALIAS
  :aliasname s
  :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
}

```

```

WARNING:  is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_s_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING:  is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_s_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING:  is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING:  Log 3: At the starting of Plan creation:

```

```

DETAIL: {PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:sortgrouprefs ( 0)
:cost.startup 0
:cost.per_tuple 0
:width 4
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 8196
:disabled_nodes 0
:startup_cost 0
:total_cost 343
:pathkeys <>
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
)
}
:startup_cost 0
:per_call_cost 368
}
{SUBPLAN
:sublinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
}
)
}
}
}
}

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 3
        :location -1
    }
    {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    }
    :location -1
    }
    :paramids (i 1)
    :plan_id 2
    :plan_name SubPlan\ 2
    :firstColType 23
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable true
    :unknownEqFalse true
    :parallel_safe true
    :setParam <>
    :parParam <>
    :args <>
    :startup_cost 363.49
    :per_call_cost 0.0025
    }
    )
    }
    ))
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :reliid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 11)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 2
    }
    )
    )
    :quals (
    {OPEXPR
    :opno 521
    :opfuncid 147
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 123
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 136
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
    )
    :location 134
    }
    )
    }
    :quals <>
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 28
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 43
    }
    :resno 1
    :resname count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>

```

```
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :subLinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :fir
stColTypepmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 0) :args ({VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :var
nullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1}) :startup_cost 0 :per_call_cost 368} {SUBPLAN :subLinkType 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresult
type 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn
3 :location -1} :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 363.49 :per_call_cost 0.0025}}) :is_pushed_down true :can_join fal
se :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_r
elids (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer
_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 521 :opf
uncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varn
osyn 2 :varattnosyn 1 :location -1} :CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 0]}) :location
-1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 2) :required_relds (
b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfami
lies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 → reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :subLinkType 0 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 2278 :fir
stColTypepmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 0) :args ({VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :va
rnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1}) :startup_cost 0 :per_call_cost 368} {SUBPLAN :subLinkType 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresul
ttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosy
n 3 :location -1} :PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramIds (i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firstColTypep
mod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 363.49 :per_call_cost 0.0025}}) :is_pushed_down true :can_join fal
se :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_
relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer
_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICTINFO :clause {OEXPR :opno 521 :opf
uncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :var
nosyn 2 :varattnosyn 1 :location -1} :CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 0]}) :location
-1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_relds (b 2) :required_relds
(b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfam
ilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) |
parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggrtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilt
er <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtrans 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggrtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfil
ter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtrans 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
DETAIL: (AGGPATH
:path.pathtype 345
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 1467642275.8996117
:path.total_cost 1467642275.9096117
:path.pathkeys <>
:subpath
{INDEXPATH
:path.pathtype 341
:parent_relds (b 2)
:path.pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1993616
:path.disabled_nodes 0
:path.startup_cost 0.4325
:path.total_cost 1467637291.8596117
:path.pathkeys <>
:info
{INDEXOPTINFO
:indexoid 17007
:reltablespace 0
:pages 16457
:tuples 6001215
:tree_height 2
:ncolumns 2
:nkeycolumns 2
:indexkeys ( 1 4)
:indexcollations ( 0 0)
:opfam ( 1976 1976)
:opcintype ( 23 23)
:sortopfam ( 1976 1976)
:reverse_sort ( false false)
:nulls_first ( false false)
:canreturn ( true true)
:relam 403
:indpred <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
```

```

:indrestrictinfo (
{RESTRICINFO
:clause
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypeMod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 368
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramIds (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypeMod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 363.49
:per_call_cost 0.0025
}
)
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b)
:right_relids (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 368
:norm_selec 0.5
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
{RESTRICINFO
:clause
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1

```

```

        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 136
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
}
:location 134
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 2)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
)
:predOK false
:unique false
:nullsnotdistinct false
:immediate true
:hypothetical false
:amcanorderbyop false
:amoptionalkey true
:amsearcharray true
:amsearchnulls true
:amhasgettupple true
:amhasgetbitmap true
:amcanparallel true
:amcanmarkpos true
}
:indexclauses (
{INDEXCLAUSE
:rinfo
{RESTRICTINFO
:clause
{OPEXPR
:opno 521
:opfncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 2)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
)
:indexquals (
{RESTRICTINFO
:clause
{OPEXPR
:opno 521
:opfncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
}
:location 134
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 0
:security_level 1
:num_base_rels 1
:clause_relids (b 2)
:required_relids (b 2)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 2)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:router_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hashoperator 0
:right_hashoperator 0
}
)
:lossy false
:indexcol 0
:indexcols <>
}
)
:indexorderbys <>
:indexorderbycols <>
:indexscandir 1
:indextotalcost 73644.6725
:indexselectivity 0.6644041354651555
}
:aggstrategy 0
:aggssplit 0
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 31
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 1467642275.8996117
:plan.total_cost 1467642275.9096117
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtransp 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno 0
:aggtransno 0
:location 43
}
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:plan.qual <>
:plan.lefttree
{INDEXSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 1467637291.8596117
:scan.plan.plan_rows 1993616
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 1
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
}
:resno 2
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
}
:resno 3
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
}
:resno 4
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location -1
}
}
:resno 5
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
}
:resno 6
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location -1
}
}
:resno 7
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY

```

```

:expr
{VAR
:varno 2
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 8
:location -1
}
:resno 8
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 10
:location -1
}
:resno 10
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 11
:location -1
}
:resno 11
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 12
:location -1
}
:resno 12
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 13
:location -1
}
:resno 13
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 14
:location -1
}
:resno 14
:resname <>

```

```

:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 15
:location -1
}
:resno 15
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 16
:location -1
}
:resno 16
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
)
:paramids (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 363.49
:per_call_cost 0.0025
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 2
:indexid 17007
:indexqual (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 123
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 136
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 134
}
)
:indexqualorig (
{OEXPR
:opno 521

```

```

:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 123
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 136
    :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
  }
)
:location 134
}
)
:indexorderby <>
:indexorderbyorig <>
:indexorderbyops <>
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 0
:aggsplit 0
:numCols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
  {RANGETBLENTTRY
    :alias
    {ALIAS
      :aliasname l_alias
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname l_alias
      :colnames ("l_orderkey" "l_linenumber")
    }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 0
    :inh false
    :relkind <>
    :rellockmode 0
    :perminfoindex 0
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTTRY
    :alias <>
    :eref
    {ALIAS
      :aliasname lineitem
      :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
    }
    :rtekind 0
    :relid 16566
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 3
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
)
:unprunableRelids (b 2 3 4)

```

```

:permInfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 13)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:resultRelations <>
:appendRelations <>
:subplans (<>
  {SEQSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 343
    :scan.plan.plan_rows 8196
    :scan.plan.plan_width 4
    :scan.plan.parallel_aware false
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 2
    :scan.plan.targetlist (
      {TARGETENTRY
        :expr
        {VAR
          :varno 4
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 4
          :varattnosyn 1
          :location -1
        }
        :resno 1
        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
  }
  :scan.plan.qual (
    {OEXPR
      :opno 674
      :opfuncid 297
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 4
          :varattno 6
          :vartype 701
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 4
          :varattnosyn 6
          :location -1
        }
        {CONST
          :consttype 701
          :consttypmod -1
          :constcollid 0
          :constlen 8
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 0 [ 0 0 0 0 0 64 -113 64 ]
        }
      )
      :location -1
    }
  )
  :scan.plan.lefttree <>
  :scan.plan.righttree <>
  :scan.plan.initPlan <>
  :scan.plan.extParam (b)
  :scan.plan.allParam (b)
  :scan.scanrelid 4
)
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16566 16541 16541)
:invalItems <>
:paramExecTypes (o 23 23)
:utilityStat <>
:stmt_location 0
:stmt_len 0
}

```

QUERY PLAN

```

Aggregate  (cost=1467642275.90..1467642275.91 rows=1 width=8) (actual time=2414.176..2414.177 rows=1.00 loops=1)
  Output: count(*)
  Buffers: shared hit=1914 read=79553
  ->  Index Scan using l_pk on public.lineitem  (cost=0.43..1467637291.86 rows=1993616 width=0) (actual time=29.140..2257.036 rows=3279256.00 loops=1)
        Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_suppkey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_extendedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_return
flag, lineitem.l_linestatus, lineitem.l_shipdate, lineitem.l_commitdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shipmode, lineitem.l_comment
        Index Cond: (lineitem.l_orderkey > 20000000)
        Filter: (ANY (lineitem.l_suppkey = (hashed SubPlan 2).col1))
        Rows Removed by Filter: 721462
        Index Searches: 1
        Buffers: shared hit=1914 read=79553
        SubPlan 2
          ->  Seq Scan on public.supplier s  (cost=0.00..343.00 rows=8196 width=4) (actual time=0.400..23.657 rows=8197.00 loops=1)
                Output: s.s_suppkey
                Filter: (s.s_acctbal > '1000')::double precision)
                Rows Removed by Filter: 1803
                Buffers: shared hit=218
Planning:
  Buffers: shared hit=132
Planning Time: 16.432 ms
Execution Time: 2422.652 ms
(20 rows)

tpch=>

```